EXPERIMENT NO. 8

Bubble Sort

Programming Language: Java

Name : A Yuvan Charan
Register No. : 192424383

Aim

To write a Java program that sorts an array of integers using the bubble sort technique and to analyse its time complexity.

Question

Sort an array of integers using the bubble sort technique. Analyze its time complexity using Big-O notation. Write the code.

Example

Input: {64, 34, 25, 12, 22, 11, 90}

Output: {11, 12, 22, 25, 34, 64, 90}

Algorithm

Step 1: Start.

Step 2: Read the array arr of size n.

Step 3: Repeat for pass i = 0 to n-2: set swapped = 0.

Step 4: For j = 0 to n-i-2, compare arr[j] and arr[j+1]; if arr[j] > arr[j+1], swap them and set swapped = 1.

Step 5: If swapped is 0 after a pass, the array is already sorted, so stop early.

Step 6: Print the sorted array.

Step 7: Stop.

Program (Java)

```
public class Main {

    static void bubbleSort(int[] arr) {
        int n = arr.length;
        for (int i = 0; i < n - 1; i++) {
            boolean swapped = false;
            for (int j = 0; j < n - i - 1; j++) {
                if (arr[j] > arr[j + 1]) {
                    int temp = arr[j];
                    arr[j] = arr[j + 1];
                    arr[j + 1] = temp;
                    swapped = true;
```

```
                }
            }
            if (!swapped)           // already sorted
                break;
        }
    }

    public static void main(String[] args) {
        int[] arr = {64, 34, 25, 12, 22, 11, 90};

        bubbleSort(arr);

        System.out.print("Output: ");
        for (int x : arr)
            System.out.print(x + " ");
        System.out.println();
    }
}
```

Output

```
Output: 11 12 22 25 34 64 90

=== Code Execution Successful ===
```

Complexity Analysis

Time Complexity: O(n^2) in the worst and average case, O(n) in the best case (already sorted array, because of the swapped flag).

Space Complexity: O(1) - sorting is done in place.

Result

Thus the program was executed successfully and the output was verified.